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Instruction Granularity as Planning Width: Formalizing and Benchmarking for Vision-Language-Action Models

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Summary

This paper formalizes instruction granularity for Vision-Language-Action (VLA) agents as planning demand and quantifies it via rule width adapted from width-based planning. It introduces Mini-BEHAVIOR-Gran, a controlled benchmark that holds tasks fixed and

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varies instruction granularity through rule-set banks and a dynamic oracle instructor, enabling a systematic study of how granularity affects performance. Experiments across three action-decoding architectures reveal a U-shaped relationship between granularity and success, an asymmetry in cross-granularity generalization (coarse→fine works better than fine→coarse), and evidence that coarse training can induce language-ignoring shortcuts, with mixed-granularity training (especially fine-leaning mixtures) mitigating this effect.

Strengths

Technical novelty and innovation

- Introduces a principled, domain-level formalization of instruction granularity as rule width grounded in iterated-width (IW) planning concepts, improving over ad hoc proxies like decomposition depth.
- Models instructions as rule sets over shared domain features, enabling a consistent, cross-task measure of planning demand and a clean disentanglement from instruction length.
- Provides a granularity-centric perspective on “language-ignoring,” linking it to the predictability of instructions from visual state via an intuitive probabilistic argument.

Experimental rigor and validation

- Designs and releases (as claimed) a controlled benchmark that isolates instruction granularity while keeping task dynamics constant, with multiple width levels per task, a dynamic oracle instructor, and template-based instruction generation.

- Evaluates across three representative action-decoding heads (autoregressive, discrete diffusion, parallel decoding) with consistent VLM backbones, strengthening the generality of findings.
- Presents analyses that decouple width from horizon (rule-call count), showing SR degrades with width even at fixed horizon, and examines failure dynamics (regression vs stagnation) aligned with width theory.

Clarity of presentation

- Provides a concrete running example and a table showing state progressions and computed widths, making the width notion intuitive for embodied tasks.
- Clearly articulates the observed U-shaped performance trend and the cross-granularity asymmetry, with supportive heatmaps and plots.

Significance of contributions

- Elevates instruction granularity to a measurable, controlled factor for diagnosing and improving multimodal grounding in VLA models, with immediate implications for training curricula and benchmark design.
- The observed trade-offs between task success and language reliance can inform future VLA training regimens aiming at both robust performance and faithful grounding.

Weaknesses

Technical limitations or concerns

- The formal link between rule width and the learning difficulty for neural VLA policies is correlational; guarantees from IW(k) in classical planning do not directly transfer to learned policies, and claims like “IW(2) is guaranteed to find a shortest-path plan” are context-dependent and may be overstated.
- The approach relies on hand-engineered feature sets Φ and hand-authored rule banks; width and instruction generation quality hinge on this symbolic design, limiting generality beyond the chosen domain.
- The language-ignoring explanation via $P(I | v)$ is plausible but not directly measured (e.g., no mutual information or predictability estimates are reported), so causal claims remain somewhat speculative.

Experimental gaps or methodological issues

- Inconsistencies in reported numbers and definitions: fine-grained instructions are described as width $w=1$ in the running example but later mapped “in most cases” to $w=0$; the ASR/ Δ SR metric and plotted deltas appear mismatched, and the text states “21% without language” for Axial-C while the figure shows $\approx 10.9\%$.
- The instruction templates are uniform and templated; the robustness of findings to natural language variability (paraphrases, noise, out-of-template phrasing) is not tested.
- The dynamic oracle instructor selects applicable rules at test time, which is an idealization; implications for real-world settings with imperfect instructors or inconsistent language are not explored.
- No comparison to non-learning or search-based baselines that use the same rule sets (e.g., symbolic planners executing rule policies) to further validate width as a predictor independent of model class.

Clarity or presentation issues

- Some notation and equation formatting artifacts and a few ambiguous statements (e.g., shortest-path guarantees for IW) may mislead readers unfamiliar with width-planning theory.
- Several figures lack complete axis labels or numeric uncertainty details in-text; some legends (e.g., attention sink/static instruction) could be clearer.

Missing related work or comparisons

- While language ignoring and VLA surveys are cited, comparisons to works that manipulate instruction conditions or design for long-horizon goal-only settings (e.g., LoHoRavens; diffusion planning/in-painting; LLM-Planner/FLARE) could be deepened to highlight how width differs from other difficulty axes.
- Broader neuro-symbolic or causal planning literature (e.g., causal prompting/PLAN) could be connected to width as a planning demand measure more explicitly.

Detailed Comments

Technical soundness evaluation

- Defining granularity as the maximum rule width over a set of instruction rules is conceptually sound and task-agnostic within a domain feature space. The running example convincingly shows how concurrent feature maintenance increases width and why serializing steps reduces width.
- However, guarantees from IW(k) are specialized; while width explains search complexity in classical planning, using it as a proxy for neural policy learning difficulty is an assumption supported empirically but not theoretically proven here.

Tightening the theoretical connection (even as a hypothesis with measurable proxies) would strengthen claims.

- The rule-set construction and well-formedness assumptions (goal-separating, terminating) are sensible for benchmarking but are strong assumptions compared to natural language instructions in the wild.

Experimental evaluation assessment

- The benchmark design is a contribution: 20 tasks, multiple width levels per task, a shared feature set, dynamic instruction delivery, and per-task quantile binning into Fine/Medium/Coarse. The explicit attempt to disentangle width from horizon by measuring rule call distributions and stratified SR is valuable.
- The main findings are well-supported: U-shaped performance curve across all three decoders, asymmetric generalization coarse→fine vs fine→coarse, and increased stagnation/regression linked to width/horizon. The width vs horizon heatmap is particularly informative.
- Attention analysis offers negative evidence (no monotonic trend of decreased language attention with coarser training), suggesting that language ignoring may not manifest at the attention-weight level; this candor is appreciated, though deeper probes (gradients, causal feature ablations) would be stronger.
- Some reported metrics/numbers are inconsistent across text and figures (e.g., Axial-C “21% without language” vs $\approx 10.9\%$ in the plot; ASR defined as $SR_{with_lang} - SR_{without_lang}$ vs deltas marked between with-language and static-goal bars). These inconsistencies undermine confidence and should be corrected.
- The choice of three decoding heads is appropriate; still, reporting variance across seeds, per-task breakdowns, and statistical tests would further bolster rigor.

Comparison with related work (using the summaries provided)

- Relative to LLM-Planner and FLARE (few-shot hierarchical planners with dynamic re-planning), this work is not a new planner but a formal analysis/benchmark focusing on instruction granularity; it complements those works by quantifying a dimension they do not control (width).
- Unlike LoHoRavens (which fixes goal-only instructions to test long-horizon reasoning), Mini-BEHAVIOR-Gran provides parallel instruction variants across widths on the same tasks, enabling controlled causal analysis of language granularity—this is a key novelty.
- Compared with diffusion planning/in-painting methods, the present work studies how the instruction structure (width) affects decoding architectures including diffusion; it is orthogonal but could benefit from a cross-benchmark validation to show if the U-shape persists in more realistic perception/control settings.
- Neuro-symbolic methods like PLAN or RimRule introduce causal or rule-based interventions; the current work's rule width measure could be positioned as a complementary diagnostic for when such interventions are most beneficial (e.g., high width).

Discussion of broader impact and significance

- Establishing instruction granularity as a measurable, controllable factor offers a new lens for diagnosing VLA grounding failures and designing curricula (e.g., fine- or coarse-skewed mixtures). The observed generalization asymmetry and language-ignoring risks are practically relevant to dataset design.
- The reliance on hand-designed features and templated instructions limits immediate transfer to complex, real-world VLA setups. Nonetheless, the conceptual framework is likely to influence future benchmark design and multimodal alignment research.

🔗 Questions for Authors

1. Can you clarify the discrepancy between the fine-grained width in the running example ($w=1$) and the experimental mapping where Fine is “in most cases” $w=0$? What exact widths correspond to Fine/Medium/Coarse across tasks?
2. The paper reports Axial-C achieving “21% without language” while Figure 8 shows $\approx 10.9\%$. Which value is correct, and how is ASR/ Δ SR computed in the figure (With vs Without, or With vs Static Goal)?
3. How sensitive are the results to language variability beyond templates? Have you tested paraphrases, lexically diverse or noisy instructions at each width level to verify that width, not phrasing, drives the observed patterns?
4. Could you provide statistical significance (e.g., across seeds) and per-task variance for the U-shaped trends and transfer asymmetry, and detail any controls on episode limits or early stopping that might affect SR?
5. How are widths validated for correctness across the 20 tasks? Are there tasks where width estimates differ significantly between your rule-based computation and empirical difficulty when controlling for horizon?
6. Have you measured a direct proxy for $P(l \mid v)$ or mutual information between language and vision to substantiate the predictability-driven language-ignoring hypothesis?
7. Do you plan to release the Mini-BEHAVIOR-Gran code, rule banks, feature definitions, and dynamic instructor to enable replication and extension?

Overall Assessment

This paper makes a timely and substantive contribution by formalizing instruction granularity as rule width and providing a benchmark that isolates this factor in VLA evaluation. The empirical findings—U-shaped performance across granularities, asymmetric generalization, and the trade-off between performance and language reliance—are insightful and likely to influence future training and evaluation practices. The methodology is thoughtfully designed and evaluated across multiple decoders, and the decoupling of width from horizon is particularly well done.

However, the work has notable limitations. It depends on hand-crafted features and rules in a simplified, grid-based environment with templated instructions, leaving questions about generality to more realistic perception and free-form language. Some inconsistencies in reported metrics and definitions need correction, and the causal interpretation of language ignoring is suggestive rather than demonstrated with direct measurements. Despite these issues, the core idea and benchmark are strong and valuable for the community.

On balance, I view this as a solid, novel contribution with clear diagnostic value and recommend acceptance after the authors address the noted inconsistencies, clarify definitions, and strengthen empirical evidence for the language-ignoring mechanism (or soften causal claims).

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